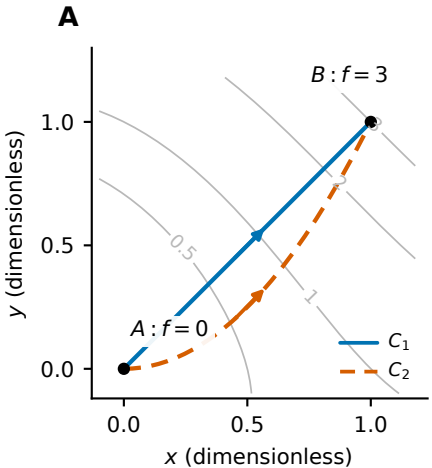
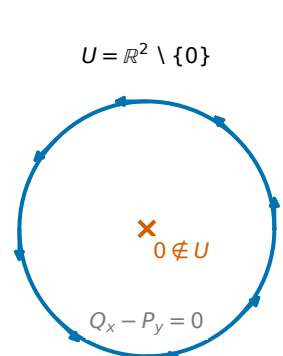


$$\int_{C_1} \nabla f \cdot d\mathbf{r} = \int_{C_2} \nabla f \cdot d\mathbf{r} = 3$$



$$\oint_C \frac{-y\,dx + x\,dy}{x^2 + y^2} = 2\pi \neq 0$$